

[Listing 11-1](#): A simple chatbot function that takes a query and returns the document that's most similar to it

[Listing 11-1](#) does not contain anything new; all of it is code that we've seen before. Now we can call the chatbot with a query about where to find something:

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```
print(chatbot('Please tell me which chapter I can go to if I want to read about
mathematics algorithms.',alldocuments))
```

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The output will tell us to go to Chapter 5:

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```
Chapter 5. Pure math, including algorithms for continued fractions and random
numbers and other mathematical ideas.
```

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Now that you've seen how the whole chatbot works, you can understand why we needed to do the normalization and vectorization. By normalizing and stemming words, we can make sure that the term *mathematics* will prompt the bot to return the Chapter 5 description, even though that exact word does not appear in it. By vectorizing, we enable the cosine similarity metric that tells us which chapter description is the best match.

We've completed our chatbot, which required stitching together a few different smallish algorithms (algorithms for normalizing, stemming, and numerically vectorizing text; an algorithm for calculating cosines of angles between vectors; and the overarching algorithm of providing chatbot answers based on query/document vector similarity). You may have noticed that we didn't manually do many of the calculations—the actual calculation of TFIDF or cosines was done by modules that we imported. In practice, you often don't need to truly understand the guts of an algorithm in order to import it and use it in your programs. This can be a blessing, in that it can accelerate our work and put amazingly sophisticated tools at our command when we need them. It can also be a curse because it causes people to misuse algorithms they don't understand; for example, an article in *Wired* magazine claimed that the misapplication of a particular financial algorithm (a method to use Gaussian copula functions to predict risks) was responsible for “kill[ing] Wall Street” and “swallow[ing] up trillions of dollars” and was a major cause of the Great Recession (<https://www.wired.com/2009/02/wp-quant/>). I encourage you to study the